INCH-POUND
MIL-DTL-16377/12C(SH)
30 June 2014
SUPERSEDING
MIL-DTL-16377/12B(SH)
2 August 1996

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SSL EQUIVALENT, GENERAL LIGHTING, 60 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 333.1, 333.2, 341.1, 341.2, 342.2, 342.3, 333.1SSL, 333.2SSL, 341.1SSL, 341.2SSL, 342.2SSL, AND 342.3SSL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances other than specified herein, see 3.5.2 of MIL-DTL-16377.

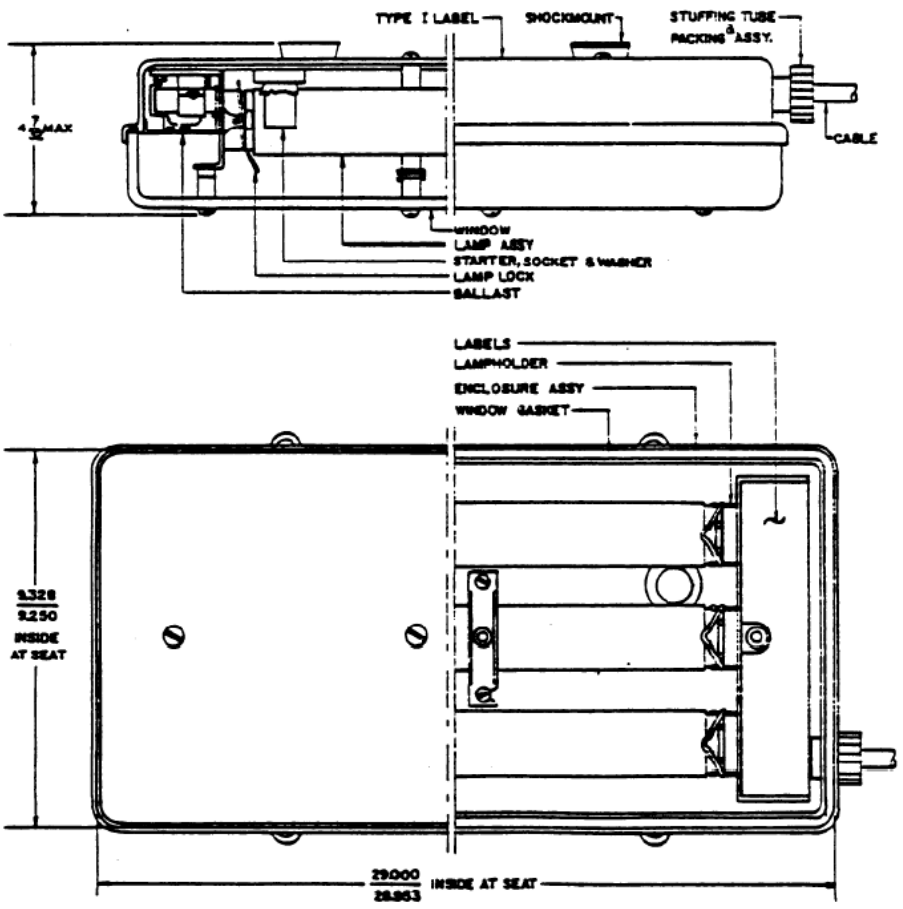


FIGURE 1. Typical view of symbols 331.1 and 342.2.

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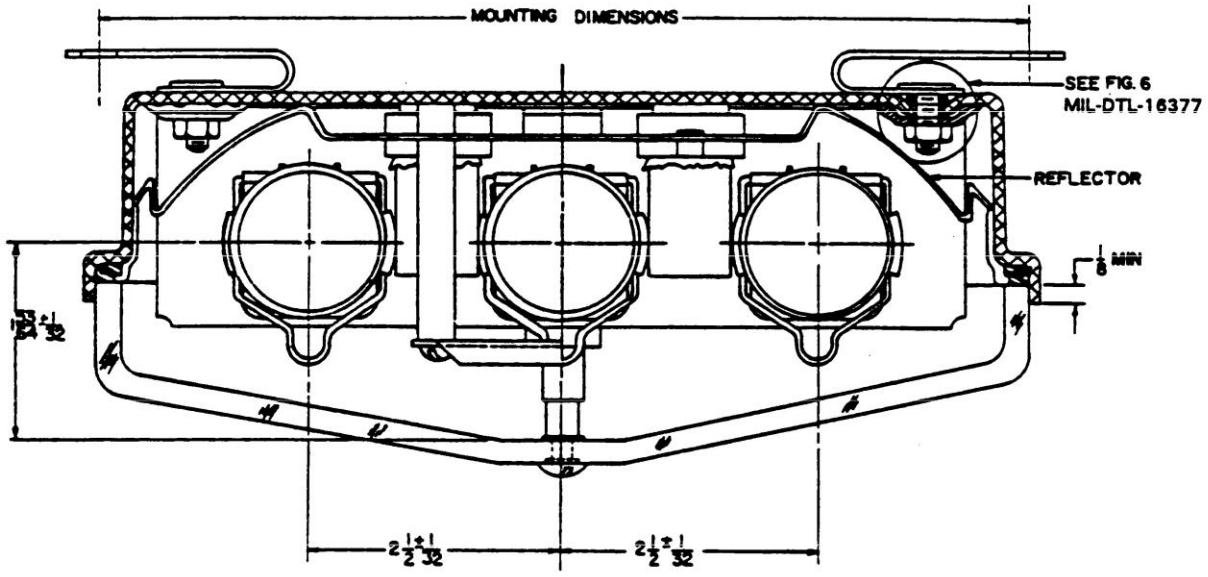


FIGURE 2. Typical section for symbols 333.1 and 342.2.

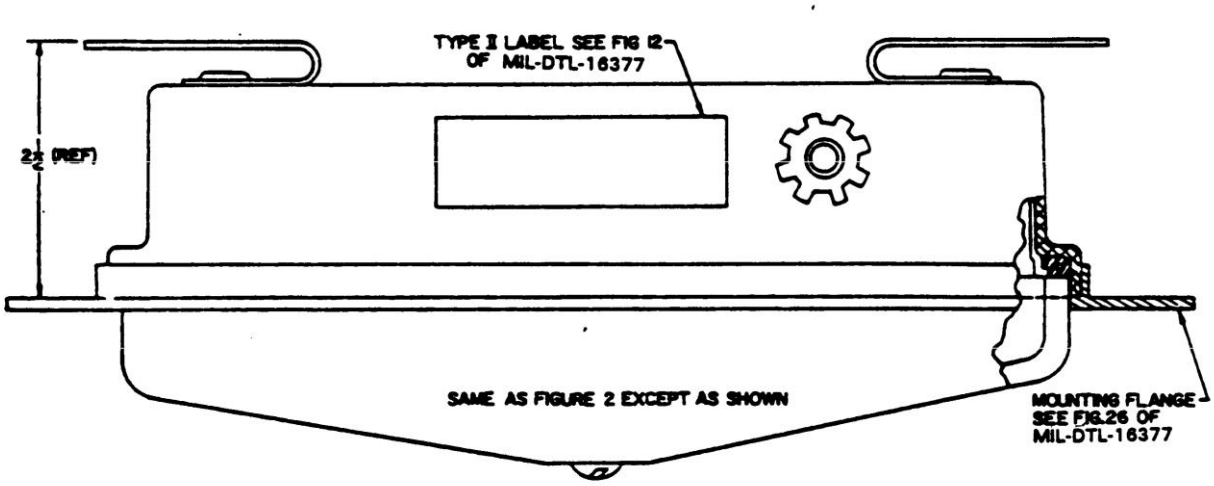


FIGURE 3. Typical section for symbols 333.2 and 342.3.

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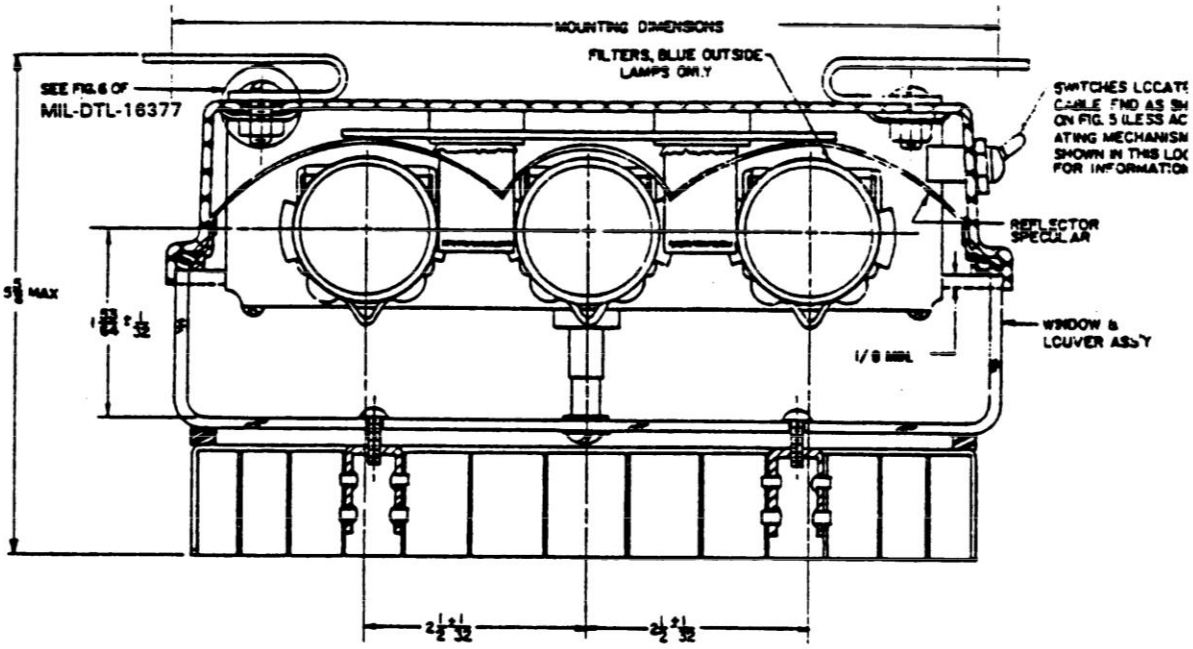


FIGURE 4. Typical section for symbol 341.1.

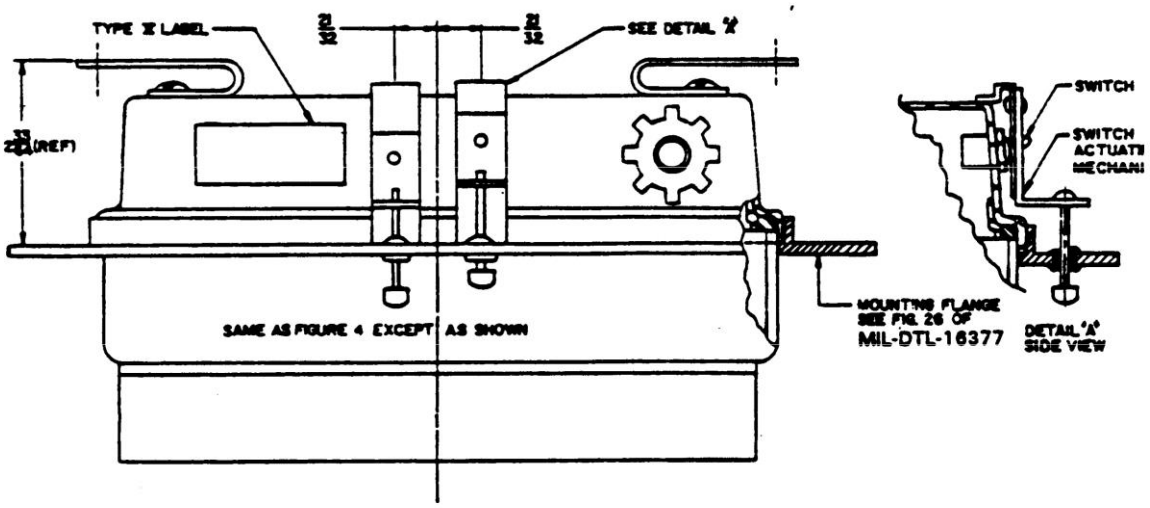


FIGURE 5. Typical section for symbol 341.2.

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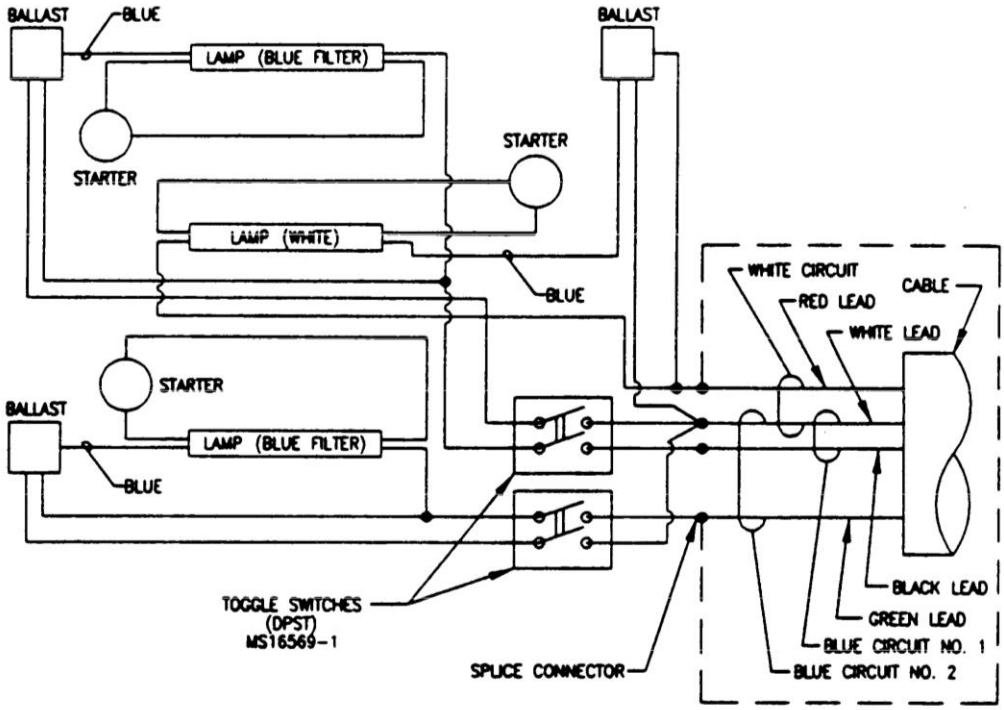
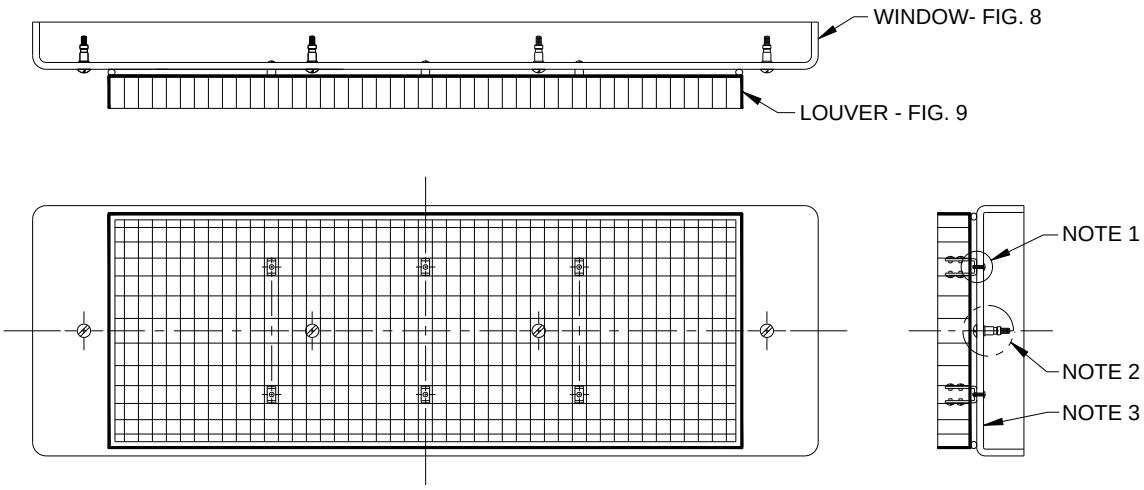


FIGURE 6. Wiring diagram for symbols 341.1 and 341.2.

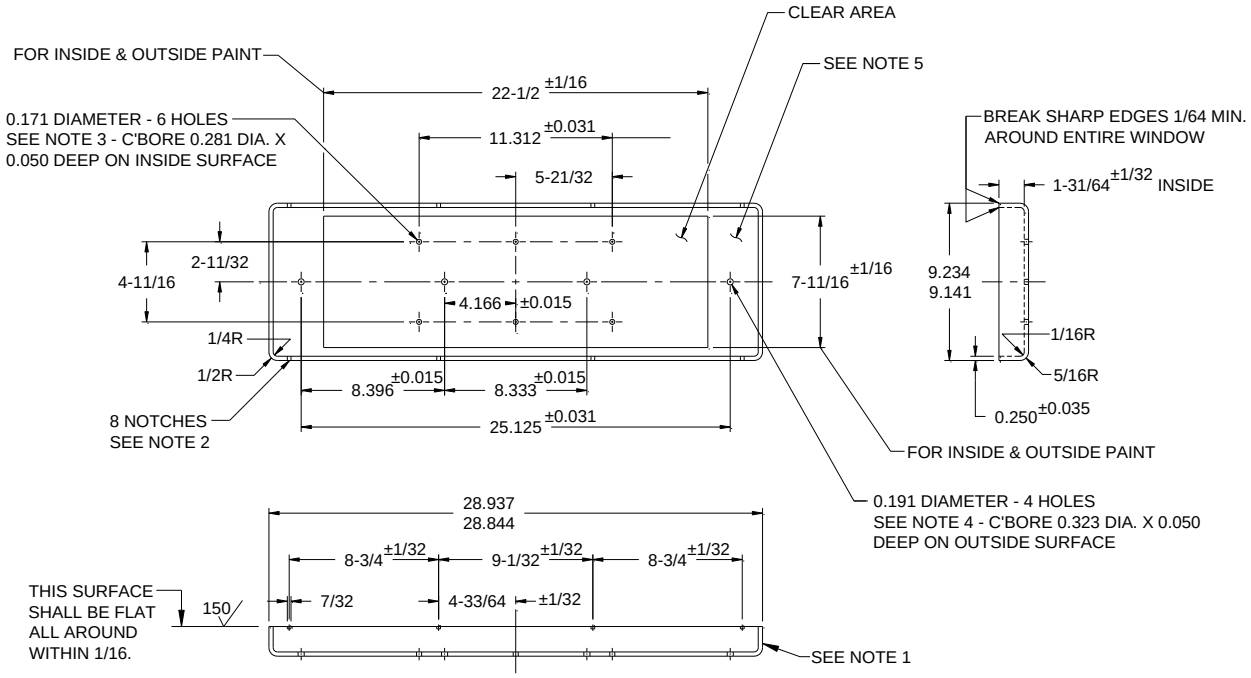


NOTES:

1. Attachment of louver to window shall be in accordance with 3.7.6.2.2 of MIL-DTL-16377.
2. Four short securing screw assemblies shall be installed in accordance with 3.7.4.3 of MIL-DTL-16377.
3. When required, one Type VI label shall be applied in accordance with 3.7.4.3 of MIL-DTL-16377.

FIGURE 7. Window assembly, clear black shielded (CIC) for symbols 341.1 and 341.2.

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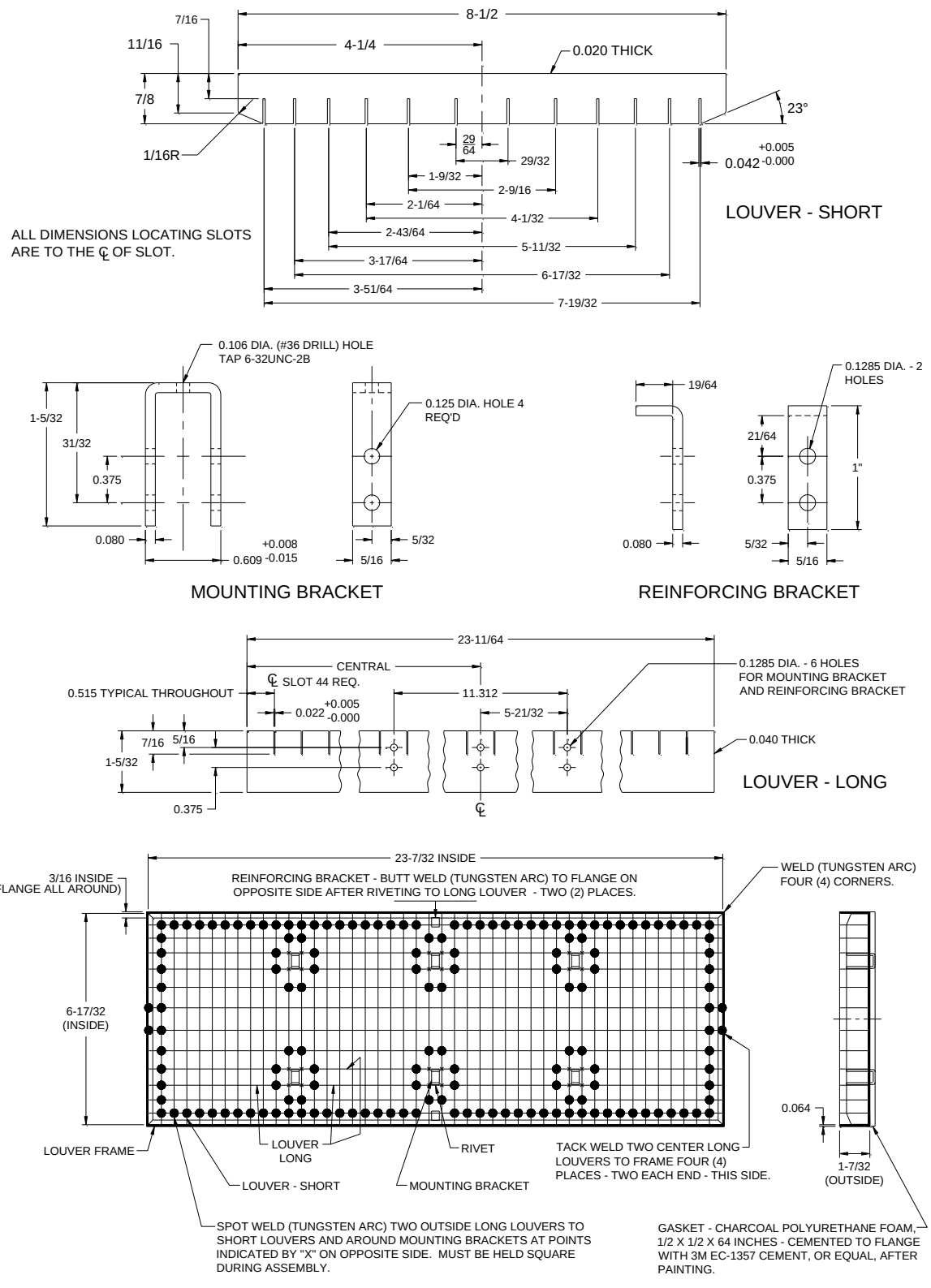


NOTES:

1. Material shall be ¼ inch thick, clear transparent acrylic plastic, Type CLEAR-1 of MIL-DTL-24191.
2. When required, notches shall be filed as shown in accordance with 3.7.4.4 of MIL-DTL-16377.
3. Break sharp edges on the outside surface of all holes.
4. Break sharp edges on the inside surface of all holes.
5. The inside surface of the window except the area indicated as “CLEAR AREA” shall be finished black. The outside surface of the window except the area indicated as “CLEAR AREA” shall be finished black. Finishing shall be in accordance with 3.8.4.1 of MIL-DTL-16377.

FIGURE 8. Window details.

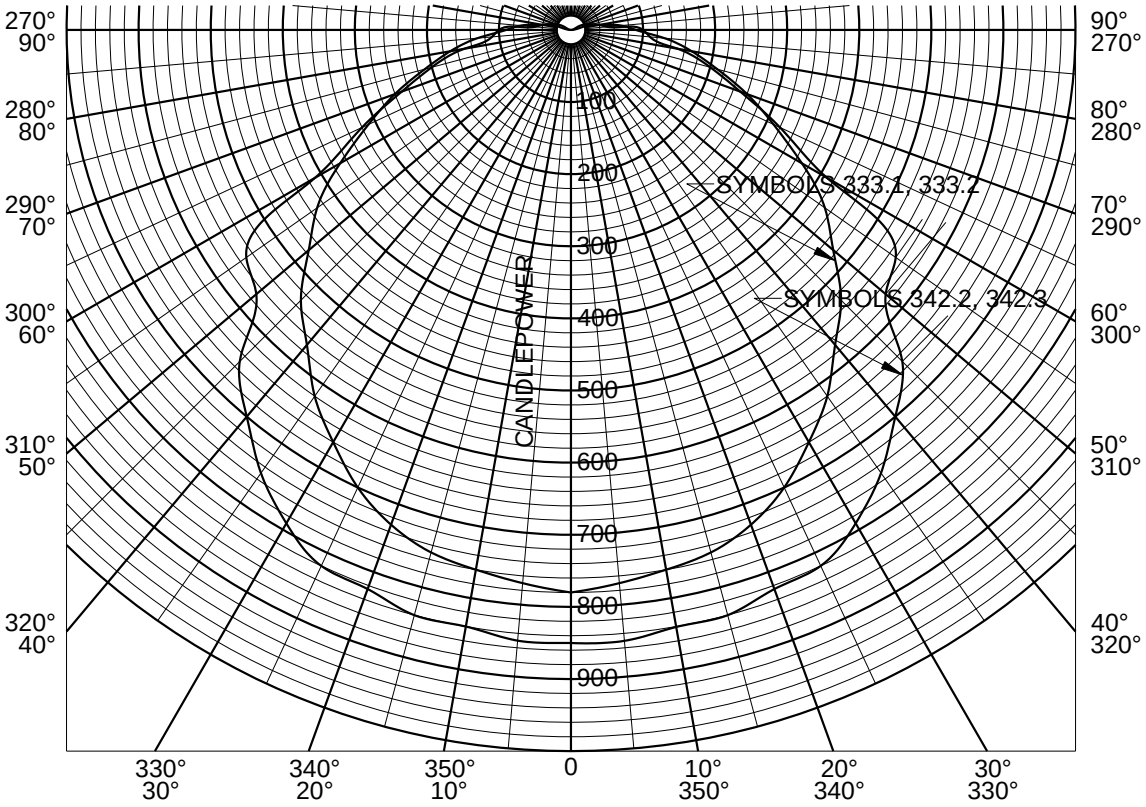
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NOTE: LOUVER ASSEMBLY SHALL BE FINISHED BLACK IN ACCORDANCE WITH 3.8.4.2 OF MIL-DTL-16377.

FIGURE 9. Louver assembly.

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PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Plane:	At right angles to long axis of luminaire through geometric center of luminaire.
Voltage:	115 volts, 60 hertz.
Ambient temperature:	77 °F (25 °C).
Fluorescent lamps:	Use lamps furnished with fixture seasoned for at least 100 hours.
Fluorescent ballast:	Use ballast furnished with fixture.
Fluorescent lamp correction:	Distribution curves based on 1,280 lumens per lamp.
SSL:	No seasoning required for LEDs.

FIGURE 10. Candlepower distribution curves for symbols 333.1, 333.2, 342.2, and 342.3.

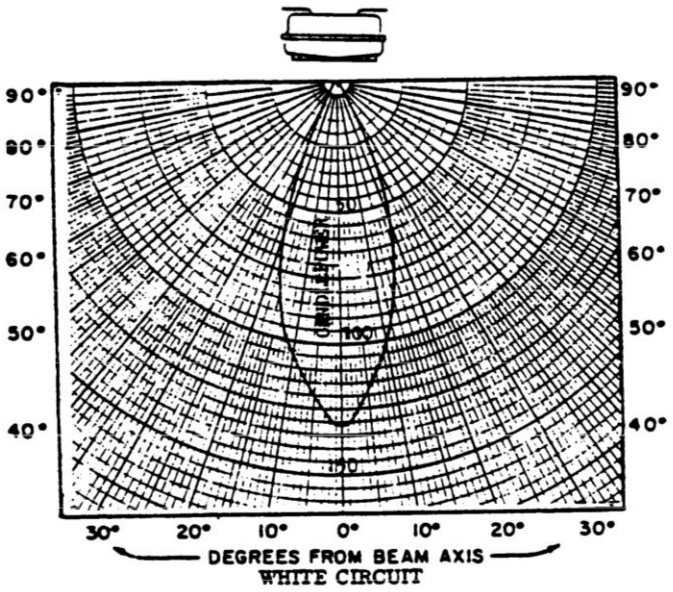
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TABLE I. Numerical candela distribution for symbols 333.1, 333.2, 342.2, and 342.3.

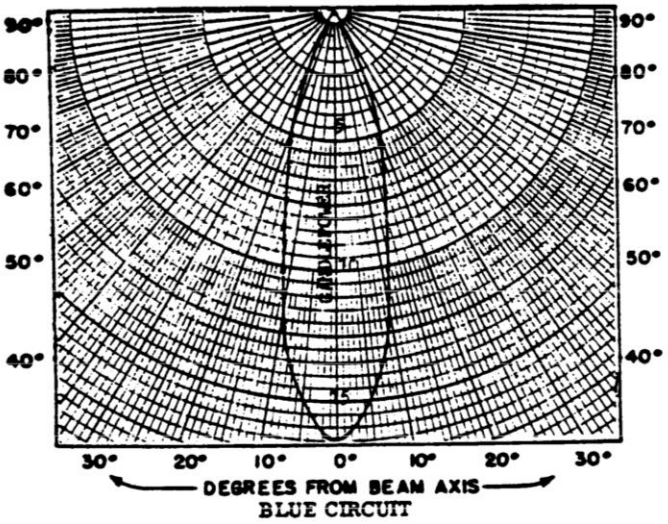
Angle	Candlepower	
	Symbol Number	
	333.1, 333.2	342.2, 342.3
0	780	850
5	770	850
10	760	840
15	750	840
20	730	825
25	700	820
30	660	790
35	620	750
40	570	700
45	530	650
50	475	570
55	430	550
60	370	410

Angle	Candlepower	
	Symbol Number	
	333.1, 333.2	342.2, 342.3
65	310	320
70	260	250
75	210	200
80	170	150
85	130	110
90	90	100
95	70	50
100	40	40
105	30	20
110	25	0
115	10	0
120	0	0
125	0	0

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Tests shall be made with center daylight lamp energized and blue filters in place on outside de-energized lamps.



Tests shall be made with center daylight lamp de-energized and blue filters in place on outside energized lamps.

PHOTOMETRIC TEST DATA:

Distance:	10 feet.
Plane:	At right angles to long axis of luminaire through geometric center of luminaire.
Voltage:	115 volts, 60 hertz.
Ambient temperature:	77 °F (25 °C).
Fluorescent lamps:	Use lamps furnished with fixtures seasoned for at least 100 hours.
Fluorescent ballast:	Use ballast furnished with fixture.
Fluorescent lamp correction:	Distribution curves based on 1,075 lumens per lamp.
SSL:	No seasoning required for LEDs.

FIGURE 11. Candlepower distribution curves for symbols 341.1 and 341.2.

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TABLE II. Numerical candela distribution for symbols 341.1 and 341.2.

Angle	Candlepower	
	Color	
	White Circuit	Blue Circuit
0	132	17
5	119	15
10	103	12
15	80	7
20	56	5
25	28	3
30	6	0
35	4	0
40	3	0
45	2	0
50	1	0
55	1	0
60	0	0

Angle	Candlepower	
	Color	
	White Circuit	Blue Circuit
65	0	0
70	0	0
75	0	0
80	0	0
85	0	0
90	0	0
95	0	0
100	0	0
105	0	0
110	0	0
115	0	0
120	0	0
125	0	0

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FLUORESCENT REQUIREMENTS:

CLASSIFICATION:

Type: I
Class: 2

SYMBOL NUMBER:

333.1: Complete lighting fixture assembly, standard mounting, with three 20-watt cool white fluorescent lamps and white diffusing window assembly.

333.2: Complete lighting fixture assembly, flush mounting, with three 20-watt cool white fluorescent lamps and white diffusing window assembly.

341.1: Complete lighting fixture, standard mounting, blue and white (Command Information Center [CIC] Lighting), clear black shielded with three 20-watt daylight lamps with blue filters over each outside lamp.

341.2: Complete lighting fixture, flush mounting, blue and white CIC Lighting, clear black shielded, with three 20-watt daylight lamps with blue filters over each outside lamp.

342.2: Complete lighting fixture assembly, standard mounting, with three 20-watt cool white fluorescent lamps and clear and white prismatic window assembly.

342.3: Complete lighting fixture assembly, flush mounting, with three 20-watt cool white fluorescent lamps and clear and white prismatic window assembly.

DESIGN:

Electrical rating: 60 watts, 120 volts, 60 hertz.

Dimensions and configuration: See figures 1, 2, 3, 4, and 5 and figure 10 of MIL-DTL-16377.

Photometric data: See figures 10 and 11 and tables I and II.

Lampholder spacing: See figure 21 of MIL-DTL-16377.

Wiring diagram: See figure 24 of MIL-DTL-16377 for Symbols 333.1, 333.2, 342.2, and 342.3 and figure 6 for Symbols 341.1 and 341.2.

Mounting dimensions:

- Symbols 333.1, 341.1, and 342.2: See figure 25 of MIL-DTL-16377.
- Symbols 333.2, 341.2, and 342.3: See figure 26 of MIL-DTL-16377.

Window assembly:

- Symbols 333.1 and 333.2: Military part number M16377/12-001, figure 27 of MIL-DTL-16377.
- Symbols 341.1 and 341.2: Military part number M16377/12-003, see figure 7.
- Symbols 342.2 and 342.3: Military part number M16377/12-004, figure 27 of MIL-DTL-16377.

Enclosure: See 3.5.11 of MIL-DTL-16377. Watertight for 333.1, 333.2, 342.2, and 342.3. Dripproof 15 degrees for 341.1 and 341.2.
Boot for toggle switch: Neoprene rubber with integral hexagon nut, enclosed or through type in accordance with MIL-DTL-5423.
Finishing for enclosures for Symbols 341.1 and 341.2: Exterior surfaces shall be finished with black baking enamel.

Cable: 10 feet of Type LSTHOF-3, in accordance with MIL-DTL-24643/3 three conductors (white, black and red) for Symbols 333.1, 333.2, 342.2, and 342.3 and LSFHOF-3 for Symbols 341.1 and 341.2.

Weight:

- Symbols 333.1 and 342.2: 15 pounds 11 ounces maximum.
- Symbols 333.2 and 342.3: 17 pounds 14 ounces maximum.
- Symbol 341.1: 19 pounds 4 ounces maximum.
- Symbol 341.2: 21 pounds 7 ounces maximum.

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Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.
b. One Type II for Symbols 333.2 and 342.3 only.
c. One Type IV. The information to be shown on the Type IV label shall be "USE THE FOLLOWING REPAIR PARTS ONLY:" followed by a list of all standard parts shown in table III and window assemblies of table VII by name and stock number only.
d. For Symbol 341.1 only: Legend on the protective cover, as shown on figure 19 of MIL-DTL-16377, shall have the cable connection portion covered with a Type III label with wiring diagram as shown by dotted box on figure 6.
e. For Symbol 341.2 only: A Type III label with wiring diagram as shown by dotted box on figure 6 shall be applied as shown on figure 5.

Standard parts: The standard parts shown in table III shall be used in the construction of the fixtures specified herein.

SSL REQUIREMENTS:

CLASSIFICATION:

Type: III
Class: 2

SYMBOL NUMBER:

333.1SSL: Complete lighting fixture assembly, standard mounting, with LED light source and white diffusing window assembly.
333.2SSL: Complete lighting fixture assembly, flush mounting, with LED light source and white diffusing window assembly.
341.1SSL: Complete lighting fixture assembly, standard mounting, blue and white CIC Lighting, clear black shielded with white and blue LED light sources.
341.2SSL: Complete lighting fixture assembly, flush mounting, blue and white CIC Lighting, clear black shielded, with white and blue LED light sources.
342.2SSL: Complete lighting fixture assembly, standard mounting, with LED light source and clear and white prismatic window assembly.
342.3SSL: Complete lighting fixture assembly, flush mounting, with LED light source and clear and white prismatic window assembly.

DESIGN:

Electrical rating: 60 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration: External dimensions cannot exceed those specified on figure 1 and figure 4.
Photometric data: See figures 10 and 11, tables I and II and 3.7.1 of MIL-DTL-16377.
Chromaticity: See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit: See 3.17 of MIL-DTL-16377.
Wiring: Shall maintain same control scheme for light output as legacy fluorescent fixture. Refer to figure 24 of MIL-DTL-16377 for Symbols 333.1SSL, 333.2SSL, 342.2SSL, 342.3SSL, and figure 6 for Symbols 341.1SSL and 341.2SSL.
Mounting dimensions:
Symbols 333.1SSL, 341.1SSL, and 342.2SSL: See figure 25 of MIL-DTL-16377.
Symbols 333.2SSL, 341.2SSL, and 342.3SSL: See figure 26 of MIL-DTL-16377.

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Window assembly:

Symbols 333.1SSL and 333.2SSL: White diffuse.

Symbols 341.1SSL and 341.2SSL: Military part number M16377/12-003, see figure 7.

Symbols 342.2SSL and 342.3SSL: Clear and white prismatic.

Enclosure: See 3.5.11 of MIL-DTL-16377. Watertight for 333.1SSL, 333.2SSL, 342.2SSL, and 342.3SSL. Dripproof 15 degrees for 341.1SSL and 341.2SSL.
Boot for toggle switch: Neoprene rubber with integral hexagon nut, enclosed or through type in accordance with MIL-DTL-5423.
Finishing for enclosures for Symbols 341.1SSL and 341.2SSL: Exterior surfaces shall be finished with black baking enamel.

Cable: 10 feet of Type LSTHOF-3 in accordance with MIL-DTL-24643/3, three conductors (white, black, and red) for Symbols 333.1SSL, 333.2SSL, 342.2SSL, and 342.3SSL and LSFHOF-3 for Symbols 341.1SSL and 341.2SSL.

Weight:

Symbols 333.1SSL and 342.2SSL: 15 pounds 11 ounces maximum.

Symbols 333.2SSL and 342.3SSL: 17 pounds 14 ounces maximum.

Symbol 341.1SSL: 19 pounds 4 ounces maximum.

Symbol 341.2SSL: 21 pounds 7 ounces maximum.

Designation and marking: The following labels shall be furnished in accordance with MIL-DTL-16377 applied as shown on figures 1 and 3:

a. Two Type I.

b. One Type II for Symbols 333.2SSL and 342.3SSL only.

c. One Type IV. The information to be shown on the Type IV label shall be “USE THE FOLLOWING REPAIR PARTS ONLY:” followed by:

(1) A list of all standard parts shown in table III and window assemblies of table VII by name and stock number only, if appropriate, or

(2) A list of available replacement parts and stock numbers, or if unavailable, manufacturer’s part numbers.

d. For Symbol 341.1SSL only: Legend on the protective cover, as shown on figure 19 of MIL-DTL-16377, shall have the cable connection portion covered with a Type III label with wiring diagram as shown by dotted box on figure 6.

e. For Symbol 341.2SSL only: One Type III label with wiring diagram as shown by dotted box on figure 6 shall be applied as shown on figure 5.

EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:

Complete fixtures: See table IV for Type I and table V for Type III.

Window assemblies: See table VI.

Conformance:

Comparison inspection: Same as first article inspection.

Conformance inspection: See MIL-DTL-16377.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table VII.

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TABLE III. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	1/ A-A-55005-1	6240-00-152-2996	For Symbols 333.1 333.2, 342.2, and 342.3
Lamp	2/ A-A-55005-3	6240-00-152-2993	For Symbols 341.1 and 341.2
Lamp lock	M16377/40-002	6250-00-399-7040	
Lampholder	UL 496	6250-00-284-0481	SUPP. SB, FIG. SB12
Startersocket	UL 496	6250-00-295-2738	SUPP. SB, FIG. SB14
Starter	NEMA ANSI C78.180	6250-00-299-2884	Commercial designation: FS2
Ballast	M16377/72-001	6250-01-465-6716	
Shock mount	M16377/41-009 M16377/41-010 M16377/41-008	6210-00-973-1747 6210-00-973-1748 6210-01-393-4302	See 3.5.16 of MIL-DTL-16377
Red filter tube	M16377/42-003	6210-00-295-2838	Not furnished, see 3.7.9 of MIL-DTL-16377
Broad band blue filter tube	M16377/42-005	6210-00-441-5138	Furnished with Symbols 341.1 and 341.2 only
Amber filter tube	M16377/42-006	6210-01-166-7184	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (1% transmission)	M16377/42-007	6210-01-498-8001	Not furnished, see 3.7.9 of MIL-DTL-16377
Gray filter tube (2.5% transmission)	M16377/42-009	6210-01-313-8168	Not furnished, see 3.7.9 of MIL-DTL-16377
Switch, toggle DPST	MS16569-1	5930-00-719-9659	For Symbols 341.1 and 341.2 only

NOTES:

1/ Industry number F20T12/CW.

2/ Industry number F20T12/D.

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TABLE IV. First article and comparison inspection for type I fixtures. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14.4	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight for Symbols 333.1, 333.2, 342.2, and 342.3.
Dripproof 15 degrees for Symbols 341.1 and 341.2.

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TABLE V. First article and comparison inspection for type III luminaires. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Power interface	3.6.18	4.8.21	
Power quality	3.6.18.1	4.8.22	
Voltage transient spike	3.6.18.2	4.8.23	
Electromagnetic interference	3.6.14	4.8.16	
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.16	4.8.3	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Shock	3.5.12	4.8.7	2/
Vibration	3.5.13	4.8.8	2/
Enclosure effectiveness	3.5.11	4.8.14	2/, 3/
Dielectric withstanding voltage	3.6.15	4.8.2	2/
Insulation resistance	3.6.14	4.8.3	2/
Noise	3.5.14	4.8.9	2/
Continuity of grounding	3.6.13.4	4.8.12	
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	
Chromaticity	3.7.4	4.8.6.5	
Correlated color temperature (CCT)	3.7.4.1.1	4.8.6.6	
Lumen maintenance	3.7.11	4.8.20	
Fail-safe circuit design	3.17	4.8.24	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Magnetic permeability	3.4.8	4.8.18	

NOTES:

1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.

2/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

3/ Watertight for Symbols 333.1SSL, 333.2SSL, 342.2SSL, and 342.3SSL.
Dripproof 15 degrees for Symbols 341.1SSL and 341.2SSL.

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TABLE VI. First article and comparison inspection for window assemblies. 1/

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.7.1	4.8.4	
Stress relief	3.8.5.3	4.8.5	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14.4	2/
Light output	3.7.1	4.8.6.1	Candlepower

NOTES:
1/ Requirement and test method paragraph numbers refer to MIL-DTL-16377.
2/ Watertight for Symbols 333.1, 333.2, 342.2, and 342.3.
Dripproof 15 degrees for Symbols 341.1 and 341.2.

TABLE VII. Part number and general information.

Military Part Number 1/ M16377/12-	Item Name	National Stock Number (NSN)	Remarks
333.1	Complete fixture, Symbol 333.1	6210-00-548-0222	2/
333.2	Complete fixture, Symbol 333.2	6210-01-135-2016	2/
341.1	Complete fixture, Symbol 341.1	6210-00-455-0068	2/
341.2	Complete fixture, Symbol 341.2	6210-01-107-2581	2/
342.2	Complete fixture, Symbol 342.2	6210-01-133-6655	
342.3	Complete fixture, Symbol 342.3	6210-01-283-0375	
333.1SSL	Complete fixture, solid state, Symbol 333.1SSL		New
333.2SSL	Complete fixture, solid state, Symbol 333.2SSL		New
341.1SSL	Complete fixture, solid state, Symbol 341.1SSL		New
341.2SSL	Complete fixture, solid state, Symbol 341.2SSL		New
342.2SSL	Complete fixture, solid state, Symbol 342.2SSL		New
342.3SSL	Complete fixture, solid state, Symbol 342.3SSL		New
001	Window assembly, white diffusing	6210-00-295-2849	3/
003	Window assembly, clear black shielded	6210-00-725-9030	4/
004	Window assembly, clear and white prismatic	6210-00-814-5914	

NOTES:
1/ Military part numbers M16377/12-342, 342.1, and 002 have been superseded by military part numbers M16377/12-342.2, 342.3, and M16377/67-005, respectively.
2 Formerly covered on NAVSHIPS Drawing 9000-S6401-74458 and MS19107.
3/ Formerly covered on MS16891-36.
4/ Formerly covered on MS16891-68.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy –SH

Preparing activity:
Navy – SH
(Project 6210-2012-026)

Review activity:
DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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